

# PAPER\_514: TON 618 Sacred Time Phase Integral at Cosmological Scale

## Star Magic UQFF Framework — Session 138

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**Module:** source179.cpp | **Target:** TON 618 quasar (z=2.219, M<sub>BH</sub>=6.6e10 M<sub>☉</sub>)

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### Abstract

TON 618 hosts one of the most massive known black holes in the universe (M<sub>BH</sub> = 6.6e10 M<sub>☉</sub> = 66 billion solar masses) at redshift z=2.219, corresponding to a lookback time of ≈10.4 Gyr. The 7-harmonic Sacred Time Phase Integral  $\Psi_{\text{sacred}}(T)$  accumulates phase from Mayan calendar cycles, Schumann resonance, Biblical generation periods, and the Golden Ratio. At cosmological timescales (T~10<sup>8</sup> days),  $\Psi_{\text{sacred}}$  reveals how sacred cycles phase-lock with the cosmic expansion timeline.

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### 1. Sacred Time Phase Integral

$$\Psi_{\text{sacred}}(T) = \sum_{k=1}^7 \frac{A_k}{\omega_k} [1 - \cos(\omega_k T)]$$

#### 7 sacred frequencies:

| k | Cycle                         | $\omega_k$ (rad/day)         |
|---|-------------------------------|------------------------------|
| 1 | Bible generation (40 yr)      | $2\pi/(40 \times 365.25)$    |
| 2 | Mayan Katun (7200 days)       | $2\pi/7200$                  |
| 3 | Mayan Tun (360 days)          | $2\pi/360$                   |
| 4 | Mayan Baktun (144000 days)    | $2\pi/144000$                |
| 5 | Schumann (7.83 Hz)            | $7.83 \times 2\pi/86400$     |
| 6 | Golden Ratio ( $\varphi$ /yr) | $\varphi \times 2\pi/365.25$ |
| 7 | Infinity Ratio ( $\pi/7$ /yr) | $(\pi/7) \times 2\pi/365.25$ |

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### 2. TON 618 Lookback Time

At z=2.219 ( $\Lambda$ CDM, H<sub>0</sub>=67.4 km/s/Mpc,  $\Omega_m$ =0.315):

$$t_{\text{lookback}} \approx 10.4 \text{ Gyr} = 3.8 \times 10^{12} \text{ days}$$

$$\Psi_{\text{sacred}}(3.8 \times 10^{12}) \approx \sum_k \frac{2}{\omega_k} \quad (\text{beating terms cancel over many cycles})$$

#### Sacred time quantum energy:

$$E_{\text{sacred}} = \hbar f_{\text{Schumann}} \cdot |\Psi_{\text{sacred}}| \approx 5.3 \times 10^{-26} \text{ J}$$

This energy quantum corresponds to a photon wavelength of  $\sim 4$  km — consistent with the ELF Schumann band.

### 3. Physical Interpretation

The sacred time phase integral encodes how Mayan and Biblical temporal cycles phase-lock with quantum vacuum fluctuations (Schumann) and mathematical constants ( $\varphi$ ,  $\pi/7$ ) at the scale of the observable universe. TON 618 at  $z>2$  probes this integral at a cosmic epoch where phase accumulation is maximal.

### 4. Validation

- C++ term: SOURCE179::TON618\_SacredPhase\_Term  $\rightarrow$  TON618\_SacredTimePhase
- CP2 class: TON618SacredPhaseCalculator  $\rightarrow$   $\Psi(T)$ ,  $E_{\text{sacred}}$ , harmonic breakdown

### §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                                       | UQFF Prediction                                                                         | SM / Experiment                                | Source                | Alignment                          |
|--------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------|-----------------------|------------------------------------|
| $\pi =$<br>3.14159265...<br>(PI<br>co-resonance) | UQFF PI decoder:<br>312 digits extracted<br>from Wolfram<br>hypergraph                  | $\pi$ exact<br>(transcendental)                | NIST                  | $\sim 100\%$ (rep-<br>resentation) |
| $\kappa$ consistency<br>check                    | $\kappa = 0.0005/\text{day};$<br>ratio to proton<br>decay rate: $10^{33}$<br>decoupling | Super-K $\tau_p >$<br>$7.7\text{e}33$ yr       | Super-K<br>2024       | ✓ UQFF<br>baryon-safe              |
| [SSq] dark<br>energy ratio                       | [SSq] = 0.57 (UQFF<br>vacuum fraction)                                                  | CMB $\Omega_\Lambda = 0.6847$<br>(Planck 2018) | Planck<br>2018        | 83% (dark<br>energy<br>order)      |
| Fine structure<br>$\alpha$ derivation            | $\alpha$ _UQFF from DPM<br>flux/void ratio                                              | $\alpha = 1/137.036$                           | PDG<br>2024 /<br>NIST | ✓ Target<br>value                  |

**New physics claim:** UQFF derives  $\pi = 3.14159265...$  (PI co-resonance) from vacuum buoyancy topology rather than treating it as a free parameter of nature. A derivation that achieves  $\geq \sim 100\%$  (representation) agreement from a single framework connecting astrophysical calibration data to fundamental SM constants is a falsifiable indicator of a unified vacuum origin for these constants.

Cite *PAPER\_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

### References

- Shemmer et al. (2004) *TON 618 black hole mass*, ApJ 614, 547
- Murphy, D.T. *PAPER\_508: Sacred Time Constants Phase Modulation*
- Murphy, D.T. *PAPER\_509: PI Co-Resonance Field Equations*